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Deep Learning and Applied AI

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1

Linear algebra

1.1 Vector spaces

A vector space V is a set with an addition and a scalar multiplication that respect some properties:

- **Commutativity**: $u + v = v + u \; \forall u, v \in V$ and also $u + v \in V$
- **Associativity:** $(u + v) + w = u + (v + w) \; \forall u, v, w \in V$ and $(ab)v = a(bv) \; \forall a, b \in \mathbb{R}$ and $av \in V$
- **Additive identity**: $\exists 0 | v + 0 = v \; \forall v \in V$
- **Additive inverse:** $\forall v \in V \; \exists w \in V | v + w = 0$
- **Multiplicative identity**: $\exists 1 \in \mathbb{R} | 1v = v \; \forall v \in V$
- **Distributive**: $a(u + v) = au + av \; \forall u, v \in V \; \forall a \in \mathbb{R}$

Example:
The set of all functions $f : [0, 1] \to \mathbb{R}$ with addition and scalar multiplication as:

$$(f + g)(x) = f(x) + g(x)$$

$$(\lambda f)(x) = \lambda f(x)$$

And additive identity and inverse as:

$$0(x) = 0$$

$$(-f)(x) = -f(x)$$

This is a vector space and all sets of functions $f : S \to \mathbb{R}$ is a vector space.

1.1.1 Base of a vector space

A base of a vector space V is a set of vectors $v_1, \ldots, v_n \in V$ that are linearly indipendent and spans V.
The span of a set of vectors is defined as:

$$\text{Span}(v_1, \ldots, v_n) = \{\alpha_1 v_1 + \cdots + \alpha_n v_n | \alpha_1, \ldots, \alpha_n \in \mathbb{R}\}$$

So, to span V and be linearly indipendent means that each vector $v \in V$ can be represented by only one linear combination of $v_1, \ldots, v_n$:

$$v = \sum_{i=1}^{n} \alpha_i v_i$$

A vector space has infinite bases, all with the same number of vectors and this number is the dimension of the vector space.

1.2 Linear maps

A linear map $T : V \to W$ is a function with some properties:

- **Additivity**: $T(u + v) = T(u) + T(v)\ \forall u, v \in V$
- **Homogeneity**: $T(\lambda v) = \lambda T(v)\ \forall v \in V\ \forall \lambda \in \mathbb{R}$

Example:
The equation:

$$y = ax + b$$

is not a linear map, because of the constant b, that in deep learning is called a bias.
This equation:

$$y = z \sin(x) + z^2 \sin(x)$$

is not a linear map in respect to z or x but is a linear map in respect to $\sin(x)$.

1.2.1 Vector space of linear maps

Fixed two vector spaces V, W, the set of all linear map $T : V \to W$ is itself a vector space with addition and scalar multiplication:

$$(S + T)(v) = S(v) + T(v)$$

$$(\lambda T)(v) = \lambda T(v)$$

The additive identity is the linear map that maps all values to the identity of the vector space W.

1.2.2 Product of linear map

Taken two linear maps $T : V \to W$ and $S : W \to Z$, their product is defined as $ST : V \to Z$:

$$(ST)(v) = S(T(v))$$

So, is the composition $S \circ T$. The product has different properties:

- **Associativity**: $(T_1 T_2) T_3 = T_1 (T_2 T_3)$
- **Identity**: $\exists I | TI = IT = T$
- **Distributive**: $(S_1 + S_2)T = S_1 T + S_2 T$

The product is not commutative in general.

1.3 Matrices

Let $T : V \to W$ be a linear map, $v_1, \ldots, v_n \in V$ the base of V and $w_1, \ldots, w_m \in W$ the base of W. The map T can be represented by a $m \times n$ matrix of values in $\mathbb{R}$:

$$T = \begin{vmatrix} T_{1,1} & \ldots & T_{1,n} \\ \vdots & \ddots & \vdots \\ T_{m,1} & \ldots & T_{m,n} \end{vmatrix}$$

The entries of the matrix are defined in such a way that:

$$T(v_j) = T_{1,j}w_1 + \cdots + T_{m,j}w_m$$

The matrix encodes how the vectors of the base are mapped, but it also maps all other vector in V:

$$T(v) = T(\sum_{\forall j} \alpha_j v_j) = \sum_{\forall j} T(\alpha_j v_j) = \sum_{\forall j} \alpha_j T(v_j)$$

The matrix representation of T depends on the base chosed.

1.3.1 Vector represented as matrix

In a vector space V with base $v_1, \ldots, v_n$ an arbitrary vector $v \in V$ can be defined as a $n \times 1$ matrix:

$$v = \begin{vmatrix} c_1 \\ \vdots \\ c_n \end{vmatrix}$$

In which the coefficient are such that:

$$v = c_1 v_1 + \cdots + c_n v_n$$

1.3.2 Matrix manipulation

A matrix is symmetric if:

$$A = A^T \implies A_{i,j} = A_{j,i} \; \forall i, j$$

For a product $A = BC$ the transpose and inverse are:

$$A^T = (BC)^T = C^T B^T$$

$$A^{-1} = (BC)^{-1} = C^{-1} B^{-1}$$

A matrix is orthogonal if:

$$A^{-1} = A^T$$

And when the matrix is orthogonal we have that $A^T A = I$.

1.3.3 Products with matrices

There a different ways in which matrices can be multiplied.
Matrix-Vector product:

$$Xy = \begin{vmatrix} | & \dots & | \\ x_1 & & x_n \\ | & \dots & | \end{vmatrix} \begin{vmatrix} y_1 \\ \vdots \\ y_n \end{vmatrix} = \sum_{j=1}^{n} y_j \begin{vmatrix} | \\ x_j \\ | \end{vmatrix}$$

Vector-Matrix product:

$$y^T X = (X^T y)^T$$

Matrix-Matrix product:

$$XY = \begin{vmatrix} - & x_1^T & - \\ \vdots & & \vdots \\ - & x_m^T & - \end{vmatrix} \begin{vmatrix} | & \dots & | \\ y_1 & & y_n \\ | & \dots & | \end{vmatrix} = \begin{vmatrix} | & \dots & | \\ Xy_1 & & Xy_n \\ | & \dots & | \end{vmatrix} = \begin{vmatrix} - & x_1^T Y & - \\ \vdots & & \vdots \\ - & x_m^T Y & - \end{vmatrix}$$

Vector-Vector product (inner):

$$x^T y = \alpha$$

Vector-Vector product (outer):

$$xy^T = \begin{vmatrix} x_1 \\ \vdots \\ x_m \end{vmatrix} \begin{vmatrix} y_1 & \dots & y_n \end{vmatrix} = \begin{vmatrix} | & \dots & | \\ y_1 x & & y_n x \\ | & \dots & | \end{vmatrix}$$

1.3.4 Matrix parameters

The trace of a matrix is the sum of the diagonal:

$$tr(A) = \sum_{\forall i} a_{i,i}$$

The trace is a linear map, since:

$$tr(A + B) = tr(A) + tr(B)$$

$$tr(\alpha A) = \alpha tr(A)$$

It is also invariant to transpose and cyclic permutations:

$$tr(A) = tr(A^T)$$

$$tr(ABC) = tr(CAB) = tr(BCA)$$

The gradient of the trace of a matrix is computed by partial derivatives:

$$\nabla tr(A) = \nabla \sum_{\forall i} a_{i,i} = \frac{\partial}{\partial a_{i,j}} \sum_{\forall i} a_{i,i}$$

The result is the identity matrix, because the derivatives are 1 only for an element in the diagonal like $a_{i,i}$.

The Frobenius norm of a matrix is:

$$||X||_F^2 = tr(XX^T) = tr(X^TX)$$

For a vector is much simple:

$$||x||_F^2 = tr(xx^T) = tr(x^Tx) = x^Tx$$

With this we can calculate the distance between two matrices:

$$||A - B||_F^2 = tr((A - B)^T(A - B)) = tr(A^TA) + tr(B^TB) - 2tr(A^TB)$$

The 1 vector (the vector with all 1) can be used to calculate sum of values of a matrix easy. To calculate the sum of values on each row:

$$A1$$

The sum of values of each column:

$$1^TA$$

The sum of all values:

$$1^TA1$$

Also we have that:

$$x^TAx = tr(x^TAx) = tr(xx^TA) = tr(XA)$$

In which $X = xx^T$ and creating this matrix is called lifting.
The permutation matrices are orthogonal matrices that are doubly stochastic and have a single 1 in every column and row:

$$\begin{vmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} 1 \\ 2 \\ 3 \end{vmatrix} = \begin{vmatrix} 2 \\ 1 \\ 3 \end{vmatrix}$$

The convex combination of two permutation matrices P, Q is another permutation matrix:

$$\alpha P + (1 - \alpha)Q = D \ \forall \alpha \in [0, 1]$$

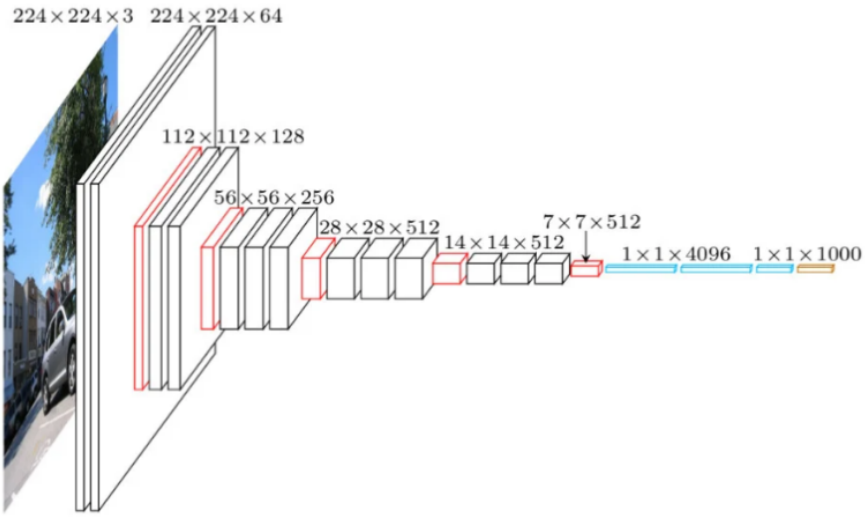
2

Deep Learning

In deep learning a model is represented by an highly parametrized model called deep neaural network. This model is a sequence of linear function with parameters Θ:

$$f_\Theta(x) = y$$

A model is a sequence of blocks, each with a predefined structure and some unknown parameters γ.



Finding the parameter values is called training and is done by minimizing a function called loss ℓ.

2.1 Convex function

A function is convex if it fullfil the **Jensen's inequality**:

$$f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha) f(y)$$

For all x, y and $\alpha \in (0, 1)$.
We also assume that f is differentiable, so we can compute the derivative $\frac{df}{dx}$ for all points. The global minimizer for x is where $\frac{df(x)}{dx} = 0$.

2.1.1 Convex functions in $\mathbb{R}^n$

If we have functions in $\mathbb{R}^n$ like the loss function:

$$\ell : \mathbb{R}^n \to \mathbb{R}$$

The derivative of this function is defined as the gradient, calculated with the partial derivatives on the vector of parameters Θ:

$$\nabla_{Theta}\ell(\Theta) = \begin{vmatrix} \frac{\partial \ell}{\partial \theta_1} \\ \vdots \\ \frac{\partial \ell}{\partial \theta_n} \end{vmatrix}$$

The loss function is convex:

$$\ell(\alpha x + (1-\alpha)y) \leq \alpha\ell(x) + (1-\alpha)\ell(y)$$

So, there is a global minimizer:

$$\nabla_\Theta \ell(\Theta) = 0 \implies \ell(x) \leq \ell(y)\ \forall y \in \mathbb{R}^n$$

2.2 Vector lenghts

In an high dimensional space $\mathbb{R}^n$, for two points $x, y \in \mathbb{R}^n$ the Euclidean distance is defined as:

$$||x-y||_2 = \sqrt{\sum_{i=1}^{d}(x_i - y_i)^2}$$

The Euclidean distance can be generalized by the L^p norm, that is defined as:

$$\delta_p(x,y) = \left(\sum_{i=1}^{d} ||x_i - y_i||^p\right)^{\frac{1}{p}}$$

The lenght (norm) of a vector is its distance from the distance:

$$||x||_2 = \sqrt{\sum_{i=1}^{d}(x_i)^2} = \sqrt{x^T x}$$

2.3 Linear regression

We take as example the simplest case, with the function:

$$f_\Theta(x_i) = ax_i + b = y_i$$

This is a linear model with 2 parameters $\Theta = \{a, b\}$ and data as pairs (x_i, y_i).
We need to choose the parameters a and b such that they minimize the **mean squared error (MSE)**, between input and predicted output:

$$\epsilon = \min_\Theta \ell(\{x_i, y_i\})$$

In which ℓ is the loss function:

$$\ell : \Theta \to \mathbb{R}$$

$$\ell_\Theta = \sum_{\forall i} (y_i - f_\Theta(x_i))^2$$

We want to minimize the loss in respect to the parameters Θ and not the data.
To minimize the parameters we want to find Θ^* minimize this function:

$$\Theta^* = \arg\min_{\Theta \in \mathbb{R}^2} (\ell(\Theta))$$

Where $\ell : \mathbb{R} \to \mathbb{R}$ is defined as:

$$\ell(a, b) = \sum_{\forall i} (y_i - (ax_i + b))^2$$

We find the solution by setting $\nabla_\Theta \ell(\Theta) = 0$:

$$\nabla_\Theta \sum_{\forall i} (y_i - (ax_i + b))^2 = \sum_{\forall i} \nabla_\Theta (y_i - (ax_i + b))^2 = \sum_{\forall i} \nabla_\Theta (y_i^2 + a^2 x_i^2 + b^2 - 2ax_i y_i - 2by_i + 2abx_i)$$

$$= \begin{vmatrix} \frac{\partial \ell}{\partial a} \\ \frac{\partial \ell}{\partial b} \end{vmatrix} = \begin{vmatrix} \sum_{\forall i} 2ax_i^2 - 2x_i y_i + 2bx_i \\ \sum_{\forall i} 2b - 2y_i + 2ax_i \end{vmatrix}$$

We set this 2 linear equation as 0:

$$\begin{vmatrix} \sum_{\forall i} 2ax_i^2 - 2x_i y_i + 2bx_i \\ \sum_{\forall i} 2b - 2y_i + 2ax_i \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \end{vmatrix}$$

2.3.1 Matrix notation

The matrix notation of the equations $y_i = ax_i + b$ can be written in 2 ways:

$$\underbrace{\begin{vmatrix} y_1 \\ \vdots \\ y_n \end{vmatrix}}_{Y} = \underbrace{\begin{vmatrix} x_1 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{vmatrix}}_{X} \underbrace{\begin{vmatrix} a \\ b \end{vmatrix}}_{\Theta} = a \begin{vmatrix} x_1 \\ \vdots \\ x_n \end{vmatrix} + b$$

In which Y is the matrix representing all the y_i and X is the matrix representing all x_i.
So, calculating the MSE is:

$$\ell(\Theta) = ||Y - X\Theta||_2^2 = (Y - X\Theta)^T (Y - X\Theta) = Y^T Y - 2Y^T X\Theta + \Theta^T X^T X\Theta$$

$$\implies \Theta = \frac{X^T Y}{X^T X} \implies \Theta = (X^T X)^{-1} X^T Y$$

2.3.2 Linear regression in $\mathbb{R}^n$

If the data points $(x_i, y_i) \in \mathbb{R}^d$ the function representing the model is:

$$y_i = Ax_i + b$$

where A is a matrix of parameters and b is a vector of bias for every dimension. Then Θ is a matrix constructed as:

$$\Theta = \begin{vmatrix} a_{1,1} & \dots & a_{1,d} & b_1 \\ \vdots & \ddots & \vdots & \vdots \\ a_{1,d} & \dots & a_{d,d} & b_d \end{vmatrix}$$

If we define the matrices X, Y as:

$$X = \begin{vmatrix} | & \dots & | \\ x_1 & & x_n \\ | & & | \\ 1 & \dots & 1 \end{vmatrix} \qquad Y = \begin{vmatrix} | & \dots & | \\ y_1 & & y_n \\ | & & | \end{vmatrix}$$

The function can be represented as:

$$\begin{vmatrix} | & \dots & | \\ y_1 & & y_n \\ | & \dots & | \end{vmatrix} = \begin{vmatrix} a_{1,1} & \dots & a_{1,d} & b_1 \\ \vdots & \ddots & \vdots & \vdots \\ a_{1,d} & \dots & a_{d,d} & b_d \end{vmatrix} \begin{vmatrix} | & \dots & | \\ x_1 & & x_n \\ | & & | \\ 1 & \dots & 1 \end{vmatrix}$$

So, we have a closed solution for Θ:

$$\Theta = (XX^T)^{-1}(XY^T)$$

2.3.3 Polynomial regression

Another type of linear regression is the polynomial regression, a regression that is linear on the paramenters but polynomial in respect to the data:

$$y_i = \sum_{j=1}^{k} a_j x_i^j + b$$

That in matrix notation is:

$$\begin{vmatrix} y_1 \\ \vdots \\ y_n \end{vmatrix} = \begin{vmatrix} x_1^k & x_1^{k-1} & \dots & x_1 & 1 \\ & & & \vdots & \\ x_n^k & x_n^{k-1} & \dots & x_n & 1 \end{vmatrix} \begin{vmatrix} a_k \\ \vdots \\ a_1 \\ b \end{vmatrix}$$

For the Stone-Weierstrass theorem:
If a function f is continuous on the interval $[a, b]$, then there exists a polynomial p such that $|f(x) - p(x)| = 0$.
We have a closed formula but only if we use the loss function as the MSE, for different loss we don't have a coled formula for minimizing the loss.

2.4 Logistic regression

We want to change the problem into a classification one, in particular a single class problem, in which the output of the model can be only 0 or 1.
So, the classification is not a continuous function.
The loss function is now not convex in a, b:

$$\ell_\Theta(\{x_i, y_i\}) = \sum_{i=1}^{n} (y_i - \sigma(ax_i + b))^2$$

In which σ is the logistic sigmoid defined as:

$$\sigma(x) = \frac{1}{1 - e^{-x}} \implies \sigma(ax_i + b) = \frac{1}{1 - e^{-(ax_i+b)}}$$

We can change the loss into two cases, making it convex:

$$\ell_\Theta(\{x_i, y_i\}) = \sum_{i=1}^{n} c(x_i, y_i) \text{ where } c(x_i, y_i) = \begin{cases} -\ln(\sigma(ax_i + b)) & y_i = 1 \\ -\ln(1 - \sigma(ax_i + b)) & y_i = 0 \end{cases}$$

This cases can be merged into a single expression:

$$c(x_i, y_i) = -y_i \ln(\sigma(ax_i + b)) - (1 - y_i) \ln(1 - \sigma(ax_i + b))$$

And the gradient is:

$$\nabla_\Theta \ell_\Theta = \nabla_\Theta \sum_{i=1}^{n} y_i \ln(\sigma(ax_i + b)) + (1 - y_i) \ln(1 - \sigma(ax_i + b)) = 0$$

To solve this we need to calculate the gradient in respect to a and to b respectively. To do that we have to calculate the derivatives of the two logarithms:

$$\nabla_\Theta \ell_\Theta = y_i \nabla_\Theta \ln(\sigma(ax_i + b)) + (1 - y_i) \nabla_\Theta \ln(1 - \sigma(ax_i + b)) = 0$$

For example we will calculate the derivatives of only $\ln(\sigma(ax_i + b))$ in respect to a applying the chain rule:

$$\frac{\partial}{\partial a} \ln(\sigma(ax_i + b)) = \frac{\partial \ln(\sigma(ax_i + b))}{\partial \sigma(ax_i + b)} \cdot \frac{\partial \sigma(ax_i + b)}{\partial (ax_i + b)} \cdot \frac{\partial (ax_i + b)}{\partial a}$$

The result of the partial derivatives is:

$$\frac{\partial}{\partial a} \ln(\sigma(ax_i + b)) = (1 - \sigma(ax_i + b))x_i$$

2.5 Gradient descent

Gradient descent is a firts-order iterative minimization algorithm. Given a loss function ℓ_Θ the algorithm works as follows:

1. Start from a point Θ^0

2. Iteratively compute:
$$\Theta^{t+1} = \Theta^t - \alpha \nabla \ell_{\Theta^t}$$
where α is called the learning rate.

The loss function ℓ_Θ must be differentiable at all points.
The gradient descent gets stuck on stationary points, points where the gradient of the loss is 0 but the point is not a global minimum or a local minimum.
The learning rate can be adaptive or follow a schedule, some example are:

- Decrease α following a parameter ρ, in different ways:
$$\alpha^{t+1} = \left(1 - \frac{t}{\rho}\right)\alpha^0 + \frac{t}{\rho}\alpha^\rho$$
$$\alpha^{t+1} = \frac{\alpha^t}{1+\rho t}$$
$$\alpha^{t+1} = \alpha^0 e^{-\rho t}$$
- Accumulate past gradients and keep moving in their direction (momentum):
$$v^{t+1} = \lambda v^t - \alpha \nabla \ell_{\Theta^t}$$
$$\Theta^{t+1} = \Theta^t + v^{t+1}$$
The steps lenght is proportional to the sequence of gradients.

We can write the formula of gradient descent in a simpler way:
$$\Theta^{t+1} = \Theta^0 - \alpha \sum_{i=0}^{t} \nabla \ell_{\Theta^t}$$
With the momentum the formula becomes:
$$\Theta^{t+1} = \Theta^0 - \alpha \sum_{i=0}^{t} \frac{1-\lambda^{t+1-i}}{1-\lambda} \nabla \ell_{\Theta^t}$$
This two formulas can be generalized for some γ_i or diagonal matrix Γ_i:
$$\Theta^{t+1} = \Theta^0 + \alpha \sum_{i=0}^{t} \gamma_i^t \nabla \ell_{\Theta^t} = \Theta^0 + \alpha \sum_{i=0}^{t} \Gamma_i^t \nabla \ell_{\Theta^t}$$

2.5.1 Stochastic gradient descent

In Deep Learning each parameter is updated to decrease the loss:
$$\theta_i = \theta_i - \alpha \frac{\partial \ell}{\partial \theta_i}$$
Calculate the loss for each training data point is a huge bottleneck, to mitigate this problem mini-batches are used. Given the training set $\mathcal{T}$ with n data points we can use a mini-batch $\mathcal{B}$ with $m < n$ data points to approximate the gradient:
$$\frac{1}{m}\sum_{i=1}^{m} \nabla \ell_\Theta(\mathcal{B}) \approx \frac{1}{n}\sum_{i=1}^{n} \nabla \ell_\Theta(\mathcal{T})$$
With the mini-batch drawn uniformly.
The stochastic gradient descent algorithm works as follows:

1. Initialize θ.
2. Pick a mini-batch $\mathcal{B}$.
3. Update the parameters:
$$\theta = \theta - \alpha \nabla \ell_\theta(\mathcal{B})$$
4. Repeat from step 2.

When the steps 2-4 cover the entire training set we have an epoch.
This change applied in stochastic gradient descent gives a significant speed-up compared to the normal gradient descent. With n training data point, d parameters and some constants related to the problem k and v, the cost of the two algorithm is:

- Gradient descent:
  - Each iteration: $O(nd)$
  - Iteration to reach the level of accuracy we want ρ: $O\left(k \log\left(\frac{1}{\rho}\right)\right)$
- Stochastic gradient descent:
  - Each iteration: $O(d)$
  - Iteration to reach the level of accuracy we want ρ: $\frac{vk^2}{\rho} + O\left(\frac{1}{\rho}\right)$

2.6 Back-propagation

Consider multiple layers of regression models, defined as:

$$g_\Theta(x) = (\sigma \circ f_{\Theta_n}) \circ \cdots \circ (\sigma \circ f_{\Theta_1})(x)$$

We call it a multi-layer perceptron or deep feed-forward neural network.
Each layer output has an hidden representation:

$$x_{l+1} = \sigma_l(W_l x_l + b_l)$$

where W_l contains the weights and b_l the bias.

Universal Approximation Theorem

For any compact set $\Omega \subset \mathbb{R}^p$, the space spanned by the function $\phi(x) = \sigma(Wx + b)$ is dense in $C(\Omega)$ for the uniform convergence.
For any continous function f there exists a q and weights U such that:

$$\left| f(x) - \sum_{k=1}^{q} u_k \phi_k(x) \right| = 0 \quad \forall x \in \Omega$$

2.6.1 Computational graphs

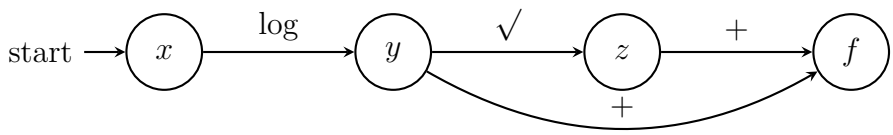
Given a generic function f the computational graph is a directed acyclic graph that represent the compuatation of f.

Example:

Given the function:

$$f(x) = \log x + \sqrt{\log x}$$

The computational graph is:



We can evaluate f by doing a forward traversal of the graph. This is done by calculating the partial derivatives of any intermediate node in respect to x.

1. $\frac{\partial x}{\partial x} = 1$
2. $\frac{\partial y}{\partial x} = \frac{\partial \log x}{\partial x} = \frac{1}{x}$
3. $\frac{\partial z}{\partial x} = \frac{\partial z}{\partial y}\frac{\partial y}{\partial x} = \frac{\partial \sqrt{y}}{\partial y}\frac{\partial y}{\partial x} = \frac{1}{2\sqrt{y}}\frac{1}{x}$
4. $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}\frac{\partial y}{\partial x} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial x} = \frac{\partial (y+z)}{\partial y}\frac{\partial y}{\partial x} + \frac{\partial (y+z)}{\partial z}\frac{\partial z}{\partial x} = \frac{1}{x} + \frac{1}{2\sqrt{y}}\frac{1}{x}$

The cost of doing this is proportional to the number of output, so for a function $f : \mathbb{R}^p \to \mathbb{R}$, the cost will be:

$$\text{compute } \nabla f(x) = p \cdot \text{compute } f(x)$$

To speed-up the process we can evaluate the function in reverse, calculating the derivatives with respect to their inner nodes.

Example:

With the same function as before:

$$f(x) = \log x + \sqrt{\log x}$$

the partial derivatives calculated in reverse are:

1. $\frac{\partial f}{\partial f} = 1$
2. $\frac{\partial f}{\partial z} = \frac{\partial (y+z)}{\partial z} = 1$
3. $\frac{\partial f}{\partial y} = \frac{\partial f}{\partial z}\frac{\partial z}{\partial y} + \frac{\partial f}{\partial y} = \frac{\partial \sqrt{y}}{\partial y} + \frac{\partial (y+z)}{\partial y} = \frac{1}{2\sqrt{y}} + 1$
4. $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}\frac{\partial y}{\partial x} = \frac{\partial f}{\partial y}\frac{\partial \log x}{\partial x} = \left(\frac{1}{2\sqrt{y}} + 1\right)\frac{1}{x} = \frac{1}{2\sqrt{y}}\frac{1}{x} + \frac{1}{x}$

The automatic differentation computing in reverse needs two steps:

1. Forward evaluation of the interior nodes, only to calculate values (no derivatives).

2. Backward pass to compute derivatives.

With loss function that always are in the form:

$$\ell : \mathbb{R}^p \to \mathbb{R}$$

where p is very high, the cost of the two approches is very different:

- Forward:
$$p \cdot \sum_{k=2}^{t-1} d_k d_{k+1}$$
- Backward:
$$\sum_{k=1}^{t-2} d_k d_{k+1}$$

The backward approach applied to deep neural network, called **back-propagation**, has a cost that is p times faster.

2.6.2 CNN

Data is often composed of hierarchical, local, shift-invariant patterns. Convolutional Neural Networks exploit this fact as a prior.

Convolution

Given two functions $f, g : [-\pi, \pi] \to \mathbb{R}$ their convolution is a function:

$$\underbrace{(f \star g)(x)}_{\text{feature map}} = \int_{-\pi}^{\pi} f(t)\, \underbrace{g(x-t)}_{\text{filter}}\, dt$$

The convolution is commutative and shift-equivariant (or translation-equivariant).
We can see convolution as the application of a linear operator $\mathcal{G}$:

$$\mathcal{G}f(x) = (f \star g)(x) = \int_{-\pi}^{\pi} f(t)g(x-t)dt$$

It's easy to show that convolution is linear:

- Homogeneity:
$$\mathcal{G}(\alpha f(x)) = \alpha \int_{-\pi}^{\pi} f(t)g(x-t)dt = \alpha \mathcal{G}f(x)$$
- Additivity:
$$\mathcal{G}(f+h)(x) = \int_{-\pi}^{\pi} f(t)g(x-t)dt + \int_{-\pi}^{\pi} h(t)g(x-t)dt = \mathcal{G}f(x) + \langle(\S)$$

Translation equivariance can be phrased as:

$$\mathcal{G}(\mathcal{T}f) = \mathcal{T}(\mathcal{G}f)$$

In the discrete setting we deal with vectors, so we define the convolution sum:

$$(f \star g)[n] = \sum_{k=-\infty}^{\infty} f[k]g[n-k]$$

The specific discretization depends on the boundary conditions.

2.6.3 Boundary conditions

In the case of images (2D) domains, for each channel the function $f : \mathbb{R}^2 \to \mathbb{R}^3$:

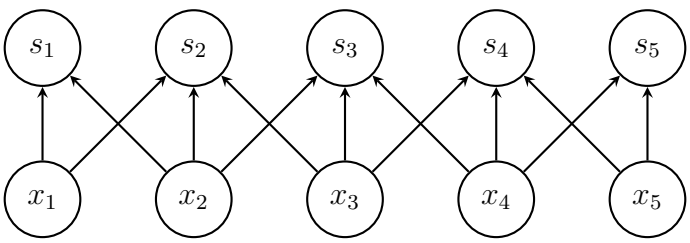
$$(f \star g)[m,n] = \sum_k \sum_l f[k,l]g[m-k, n-l]$$

The interpretation is a moving window.
We can have different boundary condition:

- **No padding**: The convolution kernel is directly applied within the boundaries of the underlying function. The result is a smaller image.
- **Full zero-padding**: The domain is enlarged and padded with zeroes. The convolution kernel is applied within the new boundaries. The result is a larger image.
- **Arbitrary zero-padding, with stride**: The domain is enlarged and padded with zeroes, but not enough to capture the boundary pixels. Also, each discrete step skips one pixel. The result is the same as no stride followed by downsampling.

2.6.4 Spatial interaction

In respect to MLP, the large matices are replaced with small local filters. Each filter has $O(1)$ cost and the weights are shared, exploiting self-similarity and shift-equivariance.
A convolutional layer is represented as:



The outgoing edges have the same weights for each input variable.

2.6.5 Pooling

To reduce the spacial dimension of the input for the next convolutional layer, pooling is used. Pooling consist in using a $m \times n$ window (overlapping if needed) and for each window return only a funtion of the values in the window (max, average, ...).

Example:

Given the following feature map:

6	8	6	3	1	0
9	13	10	5	2	0
9	14	11	6	3	0
9	13	11	6	2	0
8	13	10	5	3	0
6	7	5	3	1	0

Using a 2×2 max pooling we would have:

6	8	6	3	1	0
9	13	10	5	2	0
9	14	11	6	3	0
9	13	11	6	2	0
8	13	10	5	3	0
6	7	5	3	1	0

$\Longrightarrow$

13	10	2
14	11	3
13	10	3

Reducing the input of the successive layer from 6×6 to 3×3.